

# **INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING**

## **CHAPTER 6: Conclusion and Future Work**

by

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## 6 Conclusion and Future Work

Chapter 5 interpreted the results and judged the objectives met. The tables of Chapter 4 remain the reference.

### 6.1 Answers to the research questions

The thesis formulated EndorseRank, PageRank on a latest-allowance endorsement graph, and evaluated it against Adaptive Weighted PageRank (AWP), the transfer-graph baseline of Do, Do and Nguyen<sup>[3]</sup>, on one matched Arbitrum One wallet cohort, one spender holdout cohort and one expanded address pool over December 2025 to May 2026.

RQ1 asked whether EndorseRank and AWP produce meaningfully different wallet orderings on an identical sample. They do. The inter-method rank agreement lies strictly between unrelated and identical orderings with an interval that excludes both extremes (Table 4.3), and the rank scatter shows substantial pairwise disagreement. Hypothesis H1 is supported: authorisation graphs and payment graphs are not interchangeable reputation substrates.

RQ2 asked which method aligns more strongly with contemporaneous allowance checks and with contemporaneous transfer checks. The answer is construct-specific. AWP aligns more strongly with transfer proxies and EndorseRank with allowance proxies, and the two between-method paired differences exclude zero in opposite directions (Table 4.4). H2 is supported in that form. The within-method differences, EndorseRank’s allowance alignment against its own transfer alignment, did not reach significance, so the thesis does not claim that EndorseRank favours one signal over the other inside its own ranking.

RQ3 asked whether EndorseRank is computationally more efficient than AWP on the same matched sample. It is, on wall-clock time, memory and iterations, and the advantage is attributable to the

endorsement graph having about one ninth of the transfer graph’s edges under a single timing protocol (Table 4.1). H3 is supported. The node-count sensitivity on the expanded pool confirms the ratio across staged node counts but, because the second-tier extraction did not complete, does not test edge growth (Table 4.2).

RQ4 asked whether scores frozen at  $t_1$  predict new owner–spender approvals at  $t_2$  on inbound spenders more strongly than AWP and more informatively than the same-window allowance degree. The first holds and the second does not (Tables 4.13 and 4.14). EndorseRank’s interval excludes zero and its paired difference to AWP lies far above zero, and the labels post-date every scored event; but the spender’s raw in-approve degree at  $t_1$  predicts the same label better than EndorseRank, and the transfer in-degree predicts new senders better than AWP. Propagation separates the two constructs out of window; it does not add predictive value over the local count for these labels. RQ4 replaced the trading-success comparison of the research proposal, which was withdrawn as circular (Section 2.8); the holdout tests temporal predictive validity of the endorsement construct relative to the transfer construct, not credit-risk validity.

RQ5 asked whether a coupled PageRank over both edge types ranks wallets better than either single-layer method. It does not. Under a rule fixed before the run, C-PR at every tested  $\lambda$  lies between EndorseRank and AWP on both holdout labels and on the same-window families, and the seeded ablation reproduces AWP (Section 4.7; Table 4.14). The allowance edge contributes through propagation over it, not through seeding, and mixing it with transfer edges at the transition level dilutes it.

Claims C1–C5 (Section 4.9) describe an efficiency–alignment trade-off with two bounds. EndorseRank has lower PageRank cost, allowance-construct alignment and out-of-window separation from AWP on future approvals. AWP has stronger transfer and longitudinal-stability alignment. Neither PageRank beats its own raw degree out of window, and coupling the two edge types in one walk beats neither single-layer method. Which to deploy depends on whether the operator’s question is about authorisation or about flow, and the two scores should be kept separate.

## 6.2 Original contribution to knowledge

Section 1.9 listed four contributions. In the terms of the literature reviewed in Chapter 2, the following is established.

Before this work, the graph-reputation literature on public ledgers ranked wallets on transfers <sup>[3],[4]</sup>, on transaction-network risk propagation <sup>[12]</sup> or on explicit ratings and stake <sup>[25],[26]</sup>, and it validated those rankings against statistics drawn from the same input data. The ERC-20 allowance, the one on-chain act that records a deliberate delegation of spending authority <sup>[18]</sup>, had not been used as a reputation edge, and no same-cohort comparison with confidence intervals, difference tests and a temporal holdout existed for any pair of on-chain reputation methods. This thesis supplies that edge construct and that comparison. For the two single-layer methods PageRank is used unchanged <sup>[15]</sup>; the coupled operator changes the walk itself and is shown, under a pre-registered rule, not to improve on its end points. The knowledge claim is that the choice of edge determines what an on-chain reputation score measures, that this can be shown with quantified uncertainty on one cohort, that the allowance edge yields a score which is cheaper to refresh, distinct from the transfer-based score, aligned with its own construct and separated from the transfer score on later endorsement, while weaker than the transfer-based score on flow and stability, and that two further steps a reader might expect do not pay on this data: propagation does not out-predict the raw degree it smooths on the construct's own future growth, and a convex mixture of the two edge types does not out-rank the better single layer. The construct-validity framing <sup>[22],[33]</sup> that makes these testable statements, and the open pipeline with committed result summaries that makes them reproducible, are part of the contribution, because the auditability of decentralised credit-related scores is itself a recognised concern <sup>[2]</sup>.

The research is deliberately bounded. It does not show that on-chain reputation can substitute for collateral <sup>[1],[9]</sup>, that EndorseRank is Sybil-proof (Section 5.6) or that its results hold beyond Arbitrum One and the studied window.

## 6.3 Future work

Each item below is tied to a limitation recorded in Chapter 5.

- Completing the second-tier extraction. The expanded pool tested node count only. Extracting approval and transfer logs for the supplemental addresses would allow runtime to be measured as the edge population grows and would turn the bounded node-count statement of RQ3 into a scaling result.
- Cross-protocol and cross-chain validation. Replication on other rollups and Layer-1 networks would test whether the allowance–transfer trade-off and the holdout pattern generalise beyond Arbitrum One; the heterogeneity of DeFi venues <sup>[1],[8]</sup> makes this the main external-validity gap.
- Longer windows and a credit-linked holdout. A longer observation window would test stability of the coefficients across market regimes, and a holdout label tied to repayment or liquidation, where lending-protocol coverage of the spender cohort permits, would test whether the endorsement construct carries any credit-risk information; the present holdout does not.
- Adversarial evaluation. The Sybil cost model is analytical. Empirical red-teaming with wash approvals, coordinated spender farms and strategic revocation would quantify realised attack cost-effectiveness under live fee markets and detection heuristics <sup>[11]</sup>, and would test whether the Sybil-adjusted stability proxies remain informative under active adversaries.
- Coupling beyond a fixed transition-level mixture. The coupled operator was tested with a fixed  $\lambda$  and uniform teleportation, and it did not beat its end points. Node-dependent or learned weights, mixtures of the two stationary vectors rather than of the transition rows, and per-layer damping remain untested; any such design should be pre-registered as here, report construct-specific checks and time-split evidence with degree baselines, and preserve the auditability that distinguishes graph propagation from opaque learned scores <sup>[2],[14]</sup>.
- Beating the degree. The raw t1 degree out-predicts both PageRanks on the growth of their

own construct. Whether any propagated score adds out-of-window value over the local count, on a label that is not itself a count of arrivals, is the open question that the holdout of this thesis leaves for the field.

- Richer authorisation semantics. Adding protocol-specific delegation logs to the edge definition may improve coverage where ERC-20 approvals are sparse; the same reporting discipline applies.
- Deployment studies. Integration with risk oracles and governance tooling would test whether EndorseRank’s runtime profile sustains production refresh cadences, and controlled fee-tier or monitoring-cadence experiments would replace the adoption hypotheses of Section 5.4.4 with evidence. Collateral reduction remains a hypothesis, not a claim of this research.

## 6.4 Closing reflection

On-chain reputation does not remove the need for collateral in adversarial, pseudonymous markets [9]. It can make behavioural structure readable, and this research shows that the ranking depends on which edges are used. Transfers record economic flow; allowances record delegated authority. PageRank can rank either graph, but the same ranking cannot serve both constructs, and one walk over both graphs serves neither better than the right single graph. EndorseRank uses the authorisation graph and reports its costs and benefits, including a time-split check of later approvals in which it separates from AWP and falls short of the raw degree, two null within-method contrasts, and a pre-registered coupling test that came out negative.

A protocol or researcher who needs a fast, auditable score of who is trusted to spend has, on Arbitrum One data of the kind studied here, a usable instrument in EndorseRank. One who needs to find transfer hubs should still use AWP. Predicting trading profit from social graphs alone is outside what either method can answer.